

# OASIS: Feedback-Driven Statistics Correction for Cost-Based Query Optimizers

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**Abstract.** In large-scale analytical databases, cost-based optimizers (CBOs) depend on fresh column-level statistics, which are refreshed only by expensive **ANALYZE** passes—typically full or sampled table scans—scheduled daily or less frequently. Continuous DML workloads create a persistent *staleness gap* causing systematic selectivity estimation errors and suboptimal plans.

We present **OASIS** (**O**nline **A**daptive **S**tatistics **I**nference **S**ystem), a feedback-driven framework for *statistics correction* at the column level. Unlike plan-level feedback methods that adjust optimizer hints or join orders, OASIS operates directly on the statistics representation consumed by the CBO. It repairs stale *single-column* histograms from query execution feedback already produced during predicate evaluation, without re-scanning tables or modifying the optimizer. OASIS encodes each correction instance as a compact feature tensor and predicts corrected quantiles with a lightweight *pooled MLP* trained on simulation-derived drift scenarios. To make statistics correction portable across DBMSs, OASIS introduces *statistics-format conversion interfaces* that map engine-specific statistics layouts to a canonical full-distribution view and back, instantiated and validated on PostgreSQL-style MCV+histogram statistics.

Experiments across synthetic drift scenarios confirm that OASIS reduces Q-Error by up to 61.5% over stale priors. The same 38K-parameter checkpoint generalizes across unseen drift intensities and initial distributions within the tested drift families: at  $q=10$ , OASIS improves over the stale prior by 44%–61% across six tested distribution families. Two simple feedback-driven heuristics are decisively outperformed, confirming that learned correction is necessary. In a counterfactual end-to-end evaluation, OASIS achieves 50–55% Q-Error improvement at moderate-to-high drift, recovers 64–77% of fresh-statistics’ improvement, and resolves 73% of predicates with plan-impact risk, substantially outperforming a simulated sampled-refresh baseline. An optimizer-only EXPLAIN protocol provides suggestive evidence of the statistics→plan link on live PostgreSQL. Finally, multi-column composition experiments show an important boundary: corrected marginals propagate to downstream independence and Gaussian-copula estimators only when they remain feedback-consistent; plain OASIS gives modest/mixed E2E gains, while a lightweight OASIS-Proj calibration layer recovers 31.8%/23.1% Q-Error improvement and approaches fresh marginal inputs.

**Keywords:** Statistics correction · Cardinality estimation · Histogram calibration · Cost-based optimizer · Query feedback · Online learning

## 1 Introduction

Modern analytical query engines rely on column-level histograms to estimate predicate selectivities, which drive join-order selection and operator-algorithm choices in cost-based optimizers (CBOs). Column statistics are refreshed by periodic **ANALYZE** passes—typically involving a full or sampled table scan—that are operationally expensive and usually scheduled daily or less frequently. Meanwhile, tables undergo continuous DML workloads (inserts, deletes, updates), causing a persistent *staleness gap*: the CBO’s selectivity estimates  $\hat{s}$  diverge from true selectivities  $s^*$ , leading to suboptimal plans. For example, underestimating a base table’s cardinality can cause the optimizer to select a Nested-Loop Join that runs magnitudes slower than a Hash Join chosen after correction.

Existing approaches address this problem only partially. Frequent **ANALYZE** reruns are operationally prohibitive. Learned cardinality estimators [24,25] produce point estimates rather than corrected histograms, and require raw table data for training. Plan-level feedback methods [21,14] adjust optimizer hints but do not repair the underlying statistics—estimation errors persist across all queries referencing the affected columns. In contrast, *statistics-level* correction—the focus of this work—repairs the histogram representation itself, allowing the unmodified CBO to benefit from improved input. Self-tuning histograms [1] operate at the statistics level but require periodic data scans.

CBOs continuously generate implicit feedback during query execution: the gap between estimated selectivity  $\hat{s}$  and actual selectivity  $s^*$  is observable from operator-level runtime statistics. OASIS turns this signal into online statistics correction before the histogram reaches the CBO, without rereading raw table data. Whether this improvement translates into better plans depends on the target DBMS.

At a high level, OASIS formulates statistics correction as a regression problem over the prior histogram and a recent observation window, trains once on simulation-derived drift scenarios, and deploys the same 38K-parameter checkpoint across columns and schemas without per-table retuning. Integration requires only two lightweight hooks—a query-completion listener and a statistics-assembly intercept—and OASIS is complementary to methods targeting multi-column correlations or plan-level optimization. When composed with multi-column estimators, OASIS improves only the single-column marginal inputs; the dependence structure must still be supplied by the downstream estimator.

Our contributions are as follows:

1. **Problem formalization** (§2): We formalize statistics correction with future selectivity Q-Error as the evaluation target and use quantile regression on a fixed histogram grid as the practical surrogate optimized in practice.

2. **Feature representation and correction model** (§3): A model-agnostic feature representation encoding the prior histogram and a causal observation window, consumed by a *pooled MLP* trained on diverse drift scenarios, achieving robust performance across drift intensities.
3. **Non-invasive system integration** (§3.4, §3.5): Per-conjunct operator instrumentation and a plugin-based feedback listener, with correction injected at the statistics assembly layer via dependency injection. The design is engine-agnostic, requiring only a feedback listener and a histogram modification hook.
4. **Statistics-format conversion interface** (§3.6): A portable design pattern that maps engine-specific statistics layouts to a canonical full-distribution view and back, instantiated and validated on the representative stress case of tightly coupled MCV+histogram statistics (using PostgreSQL-style statistics objects). The same three-phase pattern (reconstruct, correct, decompose) extends to self-contained histogram formats and to other tightly coupled formats (Oracle, SQL Server). Validated with exact mass conservation, machine-precision round-trip fidelity, and sub-millisecond overhead.
5. **Systematic evaluation** (§4): Experiments across tested drift intensities ( $q \in \{1, 3, 5, 10, 15, 20, 25, 30\}$ ), initial distribution generalization, sensitivity analysis, and end-to-end evaluation through a lightweight counterfactual selectivity simulation that demonstrates per-predicate causality and plan-impact classification (73% of plan-impact-risk predicates resolved at  $q=10$ , 71% average recovery of fresh/oracle-statistics' Q-Error improvement, outperforming a simulated column-level sampled-refresh baseline). We further introduce an **optimizer-only evaluation protocol** (§4.6) that exercises the real CBO's plan selection via EXPLAIN (without ANALYZE) and validates the statistics→plan causal link without executing queries. We also add a marginal-to-joint composition study (§4.6) showing that corrected single-column marginals can improve downstream independence and Gaussian-copula estimators, but only after a lightweight feedback-consistency calibration step. Additional experiments include an observation-aggregation study (mean/max pooling vs. learned attention weights), an abstention-policy coverage-risk analysis, and comparison against simple feedback-driven heuristics that confirm the necessity of learned correction. Results are verified across three independent training seeds with tight confidence intervals.

## 2 Background and Problem Formulation

### 2.1 Column Histograms and the Staleness Gap

A CBO represents the per-column value distribution as an *equi-depth histogram* with  $B$  equal-probability buckets, defined by  $(B - 1)$  internal boundary values  $v_1, \dots, v_{B-1}$  (equivalently, quantile values at levels  $p_1, \dots, p_{B-1}$ ). Selectivity of a range predicate  $\sigma = (\text{col} < v)$  is computed as  $\hat{s} = \text{CDF}_H(v)$  via piecewise-linear CDF interpolation.

Histograms are refreshed by running **ANALYZE**—typically a full or sampled table scan—which is expensive and therefore scheduled infrequently (typically daily or less). Continuous DML operations between refreshes cause the stored histogram  $H_0$  to diverge from the true distribution  $H^*$ , inducing estimation errors.

*Format agnosticism and cross-database portability.* Our formulation operates on normalized equi-depth boundaries and is agnostic to the underlying histogram encoding. Whether the engine stores equi-depth buckets, V-optimal partitions, or streaming sketches [9], OASIS requires only that a canonical quantile/CDF view can be extracted and written back. Because many DBMSs couple the histogram with auxiliary summaries (MCV lists, density vectors), OASIS introduces a lightweight statistics-format conversion interface (§3.6) that maps engine-specific layouts to a canonical full-distribution representation and back.

## 2.2 Selectivity Feedback from Query Execution

Upon query completion, the execution engine reports per-operator input and output row counts. For a filter evaluating predicate  $\phi$  at value  $v$ , we derive the estimated selectivity  $\hat{s}$  from the CBO and the actual selectivity  $s^*$  from execution statistics. We record each observation as  $o_i = (\phi_i, v_i, v_i^{\text{upper}}, \hat{s}_i, s_i^*)$ , where  $\phi_i$  is the predicate type. These observations are collected during normal query execution without an extra table pass, adding only bounded bookkeeping (at most one record per filter conjunct).

## 2.3 Statistics-Level vs. Plan-Level Feedback

Existing feedback methods operate at the *plan level* [21,15,14], adjusting optimizer hints or plan choices based on execution history, but they do not repair the underlying stale histograms—estimation errors persist across all queries referencing the affected columns. OASIS operates at the finer *statistics level*: it corrects the histogram representation itself before it reaches the CBO. By repairing  $H_0 \rightarrow \hat{H}$  at the statistics layer, the unmodified CBO and its entire cost model benefit from improved input, and corrections propagate across all queries referencing the column. Statistics-level correction is orthogonal to plan-level methods: a system can deploy both simultaneously.

## 2.4 Problem Statement

Let  $H_0$  be the stale prior histogram captured at **ANALYZE** time and  $\mathcal{O} = \{o_1, \dots, o_K\}$  a window of  $K$  most-recent query observations for a given column. The *Q-Error* [17] of an estimate  $\hat{s}$  for true selectivity  $s^*$  is:

$$\text{QErr}(\hat{s}, s^*) = \max\left(\frac{\hat{s}}{s^*}, \frac{s^*}{\hat{s}}\right) \geq 1. \quad (1)$$

*Problem 1 (Statistics Correction).* Given the stale prior  $H_0$  and an observation window  $\mathcal{O}$ , produce corrected column statistics  $\hat{H}$  such that:

$$\mathbb{E}_\sigma[\text{QErr}(\text{CDF}_{\hat{H}}(\sigma), s_\sigma^*)] < \mathbb{E}_\sigma[\text{QErr}(\text{CDF}_{H_0}(\sigma), s_\sigma^*)],$$

where the expectation is over future predicates  $\sigma$  drawn from the workload distribution.

The solution must satisfy three operational constraints: **(D1)** no table scan (derive  $\hat{H}$  from  $H_0$  and  $\mathcal{O}$  only); **(D2)** low latency (<200 ms at query-planning time); **(D3)** graceful degradation (when  $\mathcal{O}$  is sparse,  $\hat{H} \approx H_0$ ).

*Scope.* OASIS targets *single-column histogram staleness* only. It does not repair multi-column correlations or join-level cardinality errors, and is complementary to methods that address those concerns (see Appendix for detailed scope and error-budget positioning). Corrected OASIS marginals can be passed to a downstream multi-column estimator, but the resulting estimate is only as sound as the external dependence model and the feedback consistency of the supplied marginals.

## 2.5 Problem Statement

# 3 System Design

## 3.1 System Overview

The OASIS pipeline intercepts stale prior histograms  $H_0$  before they reach the CBO. A *Feedback Collector* extracts per-conjunct observations  $o_i$  from completed queries (§3.4); a *Feature Tensorizer* encodes  $H_0$  and  $\{o_i\}$  into a fixed-size tensor  $\mathbf{x}$  (§3.2); an *Correction Model* maps  $\mathbf{x}$  to corrected quantiles  $\hat{\mathbf{v}}$ ; and  $\hat{\mathbf{v}}$  is reconstructed into the engine’s native histogram format and passed transparently to the CBO. The entire pipeline touches only two engine integration points (§3.4, §3.5), making OASIS portable across query engines without modifying the CBO or planner.

## 3.2 Feature Representation

The correction pipeline operates on normalized equi-depth boundaries. We extract  $(B-1)$  quantile values from  $H_0$  and min-max normalize to  $[0, 1]$ , decoupling the model from physical value ranges and histogram encoding. The feature tensor  $\mathbf{x}$  consists of four blocks: **(1)** Prior block ( $B-1$  dims): normalized quantile values from  $H_0$ . **(2)** Meta block (3 dims): null fraction, observation count ratio, bucket count. **(3)** Observation block ( $K \times 12$  dims): 6-dim predicate one-hot + 6 numeric features per observation, ordered causally. **(4)** Mask block ( $K$  dims): binary validity indicators. Total dimension:  $d = 220$  with  $B=10$ ,  $K=16$ .

### 3.3 OASIS Model: A Lightweight Pooled MLP

The OASIS correction model combines a compact prior encoder, attention pooling over the observation window, and a residual MLP that predicts a correction delta for the quantile vector. The prior encoder captures the coarse shape of the stale distribution; attention identifies which recent observations are most informative; the residual head updates the prior rather than regenerating the histogram from scratch. We choose attention pooling over simpler aggregation strategies because observations are heterogeneous—some are highly informative while others are redundant or noisy—and the attention mechanism learns to up-weight informative ones, which is particularly important under sparse or bursty feedback. The simple heuristic baselines (§4.3) confirm that fixed-weight aggregation is insufficient.

OASIS predicts only the internal quantile boundaries; bucket masses remain those implied by the canonical equi-depth levels. A lightweight validity projection ensures the corrected histogram is always a valid CDF summary: boundaries are clamped to  $[0, 1]$ , sorted into monotone order, and duplicates are nudged to maintain a minimum bucket width of  $10^{-6}$ . In practice, projection activates on  $<2\%$  of test predictions. Near-zero selectivities ( $s^* < 10^{-8}$ , affecting  $<0.5\%$  of test predicates) are masked during loss computation to avoid Q-Error inflation. The model is trained offline on simulation-derived ground truth, using quantile regression as a practical surrogate for downstream selectivity error. The implementation is  $\sim 550$  lines of pure Python + NumPy with  $\approx 38\text{K}$  parameters; the same checkpoint generalizes across columns without retraining. Detailed architecture and hyperparameters are provided in the supplementary materials.

### 3.4 Feedback Collection Architecture

The feedback collection design is guided by a core requirement: it must be implementable as a lightweight addition to a production engine, without EXPLAIN ANALYZE, user intervention, or disruption to existing query traffic. No CBO or planner modifications are required, but operator instrumentation and statistics hooks still need per-engine implementation ( $\approx 200\text{--}400$  lines of glue code per DBMS).

**Layer 1 (Operator instrumentation):** Filter-like operators maintain per-conjunct counters during normal query execution, updating counters on the same tuple/page reads the engine already performs. No additional table scan is introduced. **Layer 2 (Feedback listener):** Upon query completion, a listener plugin resolves each filter node’s predicate, table, and column identifiers, combines counters into observed selectivities, and flushes observation tuples to disk. One tuple is emitted per conjunct, bounding write volume by the number of conditions in the query predicate. This confines instrumentation to a small number of execution operators and requires no CBO or planner modifications. Detailed pseudocode is provided in the supplementary materials.

### 3.5 Integration and Deployment

OASIS integrates through two engine-facing touch points: **(1)** a *query-completion listener* that reads existing runtime row-count metrics and emits observation tuples per filter conjunct, and **(2)** a *statistics-assembly hook* that intercepts histogram reads immediately before they reach the CBO and substitutes corrected quantile values. Because correction is injected at the statistics assembly layer, no planner or cost-model changes are required. The query-completion listener requires per-conjunct row counts, which can be implemented as a lightweight extension accessing executor instrumentation counters maintained for EXPLAIN ANALYZE-style diagnostics; the statistics-assembly hook intercepts the histogram read path via a thin shim. Each requires modest per-engine implementation effort ( $\approx 200\text{--}400$  lines of glue code per DBMS). The feature can be toggled per session: when disabled, the unmodified stale histogram passes through with zero overhead.

### 3.6 Portable Statistics-Format Conversion Interfaces

A practical barrier to statistics correction is that many DBMSs do not expose a standalone editable histogram. PostgreSQL and Oracle couple the histogram with MCV lists; SQL Server combines it with density vectors. To make OASIS deployable, we introduce a **statistics-format conversion interface**: a three-phase translation layer that (1) reconstructs a canonical full-distribution view from the engine-native object, (2) applies OASIS correction on that canonical view, and (3) decomposes the result back to the native format. We present this as a *portable design pattern* and instantiate it for the PostgreSQL-style MCV+histogram format; for self-contained formats (MySQL, MariaDB), conversion degenerates to direct pass-through. We validate the pattern on PostgreSQL and argue its extensibility to other engines by analogy, but full cross-DBMS validation remains future work. This portability layer adds only 0.066–0.073 ms overhead (§4.2) while preserving mass conservation and machine-precision round-trip fidelity. Formal reconstruction algorithms and the detailed PostgreSQL MCV+histogram worked example are provided in the supplementary materials.

## 4 Experimental Evaluation

We organize the evaluation around four questions: **(1)** does the statistics-format conversion interface preserve correctness and low overhead; **(2)** how much does OASIS reduce histogram error as drift intensifies; **(3)** do those gains also appear in structural metrics, ablations, and initial-distribution generalization; and **(4)** do they translate into end-to-end selectivity gains with per-predicate causal evidence?

#### 4.1 Experimental Setup

Unless otherwise stated, OASIS is trained once on synthetic Gaussian compound drift and reused without per-table retuning. **Scope limitation:** all results are within tested synthetic drift families; generalization to production workloads remains open. The E2E evaluation is counterfactual rather than live, and the optimizer study covers a single PostgreSQL scenario. All experiments run on 32-core CPU, 128 GB RAM, Python 3.11. We verify stability across three training seeds (42, 123, 456) and report 95% CIs.

*Drift patterns.* We evaluate on Gaussian compound drift (insert/delete/update/null-change mutations); 6000 samples over  $q \in \{1, 3, 5, 10, 15, 20\}$  for training,  $q \in \{1, 3, 5, 10, 15, 20, 25, 30\}$  for testing. We separately test generalization across different initial distributions at  $q=10$ .

*Baselines and metrics.* We compare against the uncorrected **Stale Prior** and five feedback-driven baselines consuming the same stale histogram and feedback source: **LinInterp** (fixed 0.5 blend), **FeedAvg** (exponential moving average), **STHoles** [4], **ISOMER** [16], and **QuickSel-H** [18]. All methods use the same observation window  $K=16$  and emit corrected quantiles on the same  $B=10$  histogram grid, operating under a matched feedback budget without per-dataset retuning (see supplementary materials for baseline fairness details and QuickSel-H adaptation). Metrics include Q-Error (Eq. 1), Selectivity MAE, and Quantile MAE, averaged over 50 random test predicates per configuration. For end-to-end evaluation we use a lightweight counterfactual selectivity protocol (§4.6): the same pipeline-generated compound drift test samples are evaluated under three statistics states (stale prior, OASIS-corrected, and a simulated sampled-refresh proxy) against ground-truth post-drift quantiles. Because only the statistics input changes while the underlying data is fixed, any Q-Error difference is directly attributable to the statistics repair method. This design provides per-predicate causal evidence without requiring database engine modification.

#### 4.2 Statistics-Format Interface Validation

The PostgreSQL instantiation of the statistics-format conversion interface (§3.6) preserves total probability mass exactly: round-trip residual-quantile MAE is at machine precision (mean  $2.4 \times 10^{-17}$ , max  $9.2 \times 10^{-17}$ ) across all tested distributions. Selectivity estimates remain consistent before and after conversion (global MAE  $1.1 \times 10^{-4}$ , worst case  $< 0.39\%$ ). With PostgreSQL-like defaults (100 MCV entries, 10–20 residual bins), the full round-trip completes in 0.066–0.073 ms, and the MCV frequency threshold  $\tau_{\text{MCV}}$  provides a robust accuracy/compactness trade-off at PostgreSQL’s default ( $\approx 1\%$ ). Detailed validation results are provided in the supplementary materials.



Table 1: Q-Error comparison across drift intensities ( $\downarrow$  better). Clean rerun from the unified synthetic suite. Bold = best per row; “+ %” = improvement over the stale prior.

| $q$       | Stale Prior |   | STHoles |        | QuickSel-H |        | ISOMER       |               | OASIS (ours) |               |
|-----------|-------------|---|---------|--------|------------|--------|--------------|---------------|--------------|---------------|
|           | Q-Err       | – | Q-Err   | +%     | Q-Err      | +%     | Q-Err        | +%            | Q-Err        | +%            |
| <b>1</b>  | 1.188       | – | 1.098   | +7.5%  | 1.193      | –0.5%  | <b>1.092</b> | <b>+8.0%</b>  | 1.150        | +3.2%         |
| <b>3</b>  | 1.460       | – | 1.216   | +16.7% | 1.331      | +8.9%  | <b>1.184</b> | <b>+18.9%</b> | 1.186        | +18.8%        |
| <b>5</b>  | 1.749       | – | 1.335   | +23.7% | 1.478      | +15.5% | 1.247        | +28.7%        | <b>1.215</b> | <b>+30.6%</b> |
| <b>10</b> | 2.633       | – | 1.715   | +34.9% | 1.996      | +24.2% | 1.544        | +41.3%        | <b>1.281</b> | <b>+51.3%</b> |
| <b>15</b> | 2.995       | – | 1.769   | +40.9% | 2.053      | +31.5% | 1.649        | +45.0%        | <b>1.353</b> | <b>+54.8%</b> |
| <b>20</b> | 3.028       | – | 1.728   | +42.9% | 2.048      | +32.4% | 1.668        | +44.9%        | <b>1.276</b> | <b>+57.9%</b> |
| <b>25</b> | 3.383       | – | 1.867   | +44.8% | 2.216      | +34.5% | 1.733        | +48.8%        | <b>1.303</b> | <b>+61.5%</b> |
| <b>30</b> | 3.220       | – | 1.658   | +48.5% | 1.962      | +39.1% | 1.716        | +46.7%        | <b>1.268</b> | <b>+60.6%</b> |

Table 2: Simple heuristic baselines vs. OASIS at selected drift intensities. Results are averaged over three independent training seeds (42, 123, 456) with a different evaluation protocol than the main comparison in Table 1; absolute Q-Error values are therefore not directly comparable across the two tables. Both heuristics consume the same observation window  $K=16$ . FeedAvg worsens estimation at all drift levels.

| $q$       | Q-Error ( $\downarrow$ better) |           |         |              |
|-----------|--------------------------------|-----------|---------|--------------|
|           | Stale                          | LinInterp | FeedAvg | OASIS        |
| <b>5</b>  | 1.791                          | 1.628     | 1.835   | <b>1.238</b> |
| <b>10</b> | 2.471                          | 2.118     | 2.565   | <b>1.255</b> |
| <b>20</b> | 3.370                          | 2.749     | 3.553   | <b>1.296</b> |
| <b>30</b> | 3.500                          | 2.823     | 3.690   | <b>1.345</b> |

### 4.3 Main Results: Q-Error vs. Drift Intensity

Table 1 and Figure 1 evaluate OASIS on Gaussian compound drift, including unseen intensities  $q \in \{25, 30\}$ . The Stale Prior rises from 1.19 at  $q=1$  to above 3.0 once  $q \geq 15$ , peaking at 3.38 on the unseen  $q=25$  setting. Table 2 isolates the comparison against simple heuristic baselines that consume the same feedback window: OASIS outperforms both by a wide margin at moderate-to-high drift, and FeedAvg actually *worsens* estimation at all tested drift levels.

*In-distribution performance and unseen intensities.* At  $q=1$ , the ISOMER baseline is strongest, indicating that very mild drift can already be captured by consistency-based local refinement while OASIS remains deliberately conservative under desideratum **D3**. At  $q=3$ , OASIS nearly ties the strongest classical refiner but ISOMER remains best. From  $q \geq 5$ , OASIS is the best method at every drift level under the same  $K=16$  feedback budget, and the gap widens as the histogram becomes more stale. At  $q=10$ –30, OASIS reduces Q-Error by 51.3%–61.5% versus the Stale Prior, and this advantage persists on the unseen test intensities  $q=25$  and 30. The strongest classical competitor alternates between

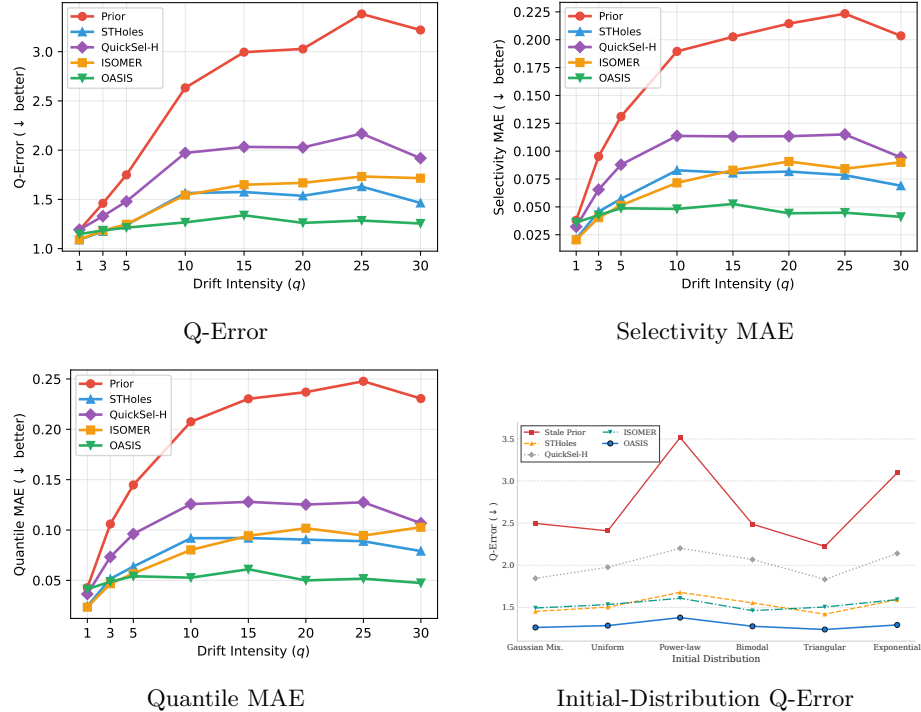


Fig. 1: Correction performance across drift intensities and initial distributions. The first three panels report Gaussian compound-drift results for  $K=16$  and  $B=10$ ; the fourth panel summarizes Q-Error under unseen initial distributions at  $q=10$ . OASIS achieves up to 61.5% Q-Error reduction over the stale prior while remaining robust across different starting distributions.

hierarchical STHoles and ISOMER, but neither matches OASIS once drift is no longer near-stale.

*Learned correction is necessary.* Both simple heuristic baselines are decisively outperformed by OASIS. **LinInterp** achieves only 14%–19% improvement at  $q \geq 10$  (versus OASIS’s 51%–61%), confirming that fixed-weight linear blending of prior and feedback is insufficient when drift is non-trivial. **FeedAvg** actually *worsens* estimation at all drift levels (negative improvement), demonstrating that naive temporal smoothing without per-region adaptation is counterproductive. Critically, OASIS also outperforms the three *published classical feedback-driven methods*: hierarchical STHoles, ISOMER, and QuickSel-H all consume the same feedback budget but fall short of OASIS at  $q \geq 5$  (Table 1), confirming that the advantage of learned correction persists against the strongest available non-neural baselines in this setting. Together, these results confirm that a learned non-linear correction model is beneficial under non-trivial drift. They do not, by themselves, isolate attention as the decisive architectural component; a controlled ablation

Table 3: Q-Error across different initial distributions at  $q=10$ . A model trained only on Gaussian-mixture initial states remains best on all tested distributions.

| Initial Distribution | Stale | STHoles | QuickSel-H | ISOMER | OASIS        |
|----------------------|-------|---------|------------|--------|--------------|
| Gaussian Mixture     | 2.497 | 1.453   | 1.845      | 1.493  | <b>1.262</b> |
| Uniform              | 2.409 | 1.503   | 1.977      | 1.534  | <b>1.284</b> |
| Skewed (Power-law)   | 3.517 | 1.679   | 2.202      | 1.609  | <b>1.379</b> |
| Bimodal              | 2.488 | 1.554   | 2.069      | 1.462  | <b>1.276</b> |
| Triangular           | 2.225 | 1.420   | 1.833      | 1.505  | <b>1.238</b> |
| Exponential          | 3.097 | 1.592   | 2.141      | 1.593  | <b>1.292</b> |

against learned mean/max pooling within the same architecture remains future work.

*Seed stability.* Across three independent seeds (42, 123, 456), OASIS exhibits the tightest cross-seed variance: at  $q=10$ , the 95% CI for Q-Error is  $1.255 \pm 0.006$ , compared to  $1.445 \pm 0.082$  for ISOMER and  $1.482 \pm 0.081$  for STHoles. The method ranking is consistent across all seeds at  $q \geq 10$ .

*Safe deployment under mild drift.* At  $q=1$ , OASIS’s Q-Error (1.150) is closer to the stale prior (1.188) than to the best classical method (ISOMER at 1.092), consistent with **D3**: the residual head predicts small corrections when feedback is uninformative.

*Predicate coverage sensitivity.* OASIS correction quality depends on the observation window covering the value range relevant to future predicates. We artificially restrict observations to a “hot region” of width  $w$  centered at the domain midpoint and measure Q-Error degradation at  $q=10$ . At  $w=0.7$  (70% coverage), Q-Error degrades by only 3%; at  $w=0.3$ , degradation reaches 40% but OASIS still outperforms the stale prior by 26%. Even at  $w=0.15$ , OASIS outperforms the stale prior. The abstention policy (§4.6) provides a natural guard for sparse-feedback scenarios. See supplementary materials for full results.

#### 4.4 Generalization Across Initial Distributions

We vary the initial table shape to test whether generalization depends on the pre-drift distribution used during training. Applying the same mutation process to six different starting histograms (uniform, skewed/power-law, bimodal, triangular, exponential, and the training Gaussian mixture) yields different post-drift distributions.

Table 3 shows consistent 44%–61% Q-Error reduction across all tested initial distributions, even though training uses only Gaussian-mixture initial states. OASIS is best on every evaluated shape, supporting the claim that the normalized quantile representation transfers beyond the training distribution.

#### 4.5 Structural Accuracy

Table 4 complements Q-Error with Quantile MAE and Selectivity MAE. OASIS is the best method on both metrics for every drift level  $q \geq 5$ . At  $q=10$ , Quantile MAE drops from 0.208 to **0.055** and Selectivity MAE from 0.190 to **0.049**, substantially below both ISOMER (0.080 / 0.072) and STHoles (0.115 / 0.105), indicating that OASIS reconstructs histogram shape rather than merely matching observed predicates.

Table 4: Quantile MAE and Selectivity MAE across drift intensities from the clean synthetic rerun. Bold denotes best method per metric.

| $q$       | Quantile MAE ( $\downarrow$ ) |         |        |              |              | Selectivity MAE ( $\downarrow$ ) |         |        |              |              |
|-----------|-------------------------------|---------|--------|--------------|--------------|----------------------------------|---------|--------|--------------|--------------|
|           | Prior                         | STHoles | QSel-H | ISOMER       | OASIS        | Prior                            | STHoles | QSel-H | ISOMER       | OASIS        |
| <b>1</b>  | 0.043                         | 0.027   | 0.036  | <b>0.023</b> | 0.041        | 0.038                            | 0.024   | 0.031  | <b>0.020</b> | 0.035        |
| <b>3</b>  | 0.106                         | 0.064   | 0.071  | <b>0.047</b> | 0.048        | 0.095                            | 0.057   | 0.064  | <b>0.040</b> | 0.042        |
| <b>5</b>  | 0.145                         | 0.085   | 0.094  | 0.057        | <b>0.054</b> | 0.131                            | 0.077   | 0.085  | 0.051        | <b>0.048</b> |
| <b>10</b> | 0.208                         | 0.115   | 0.126  | 0.080        | <b>0.055</b> | 0.190                            | 0.105   | 0.114  | 0.072        | <b>0.049</b> |
| <b>15</b> | 0.230                         | 0.119   | 0.128  | 0.094        | <b>0.062</b> | 0.203                            | 0.104   | 0.113  | 0.083        | <b>0.054</b> |
| <b>20</b> | 0.237                         | 0.115   | 0.127  | 0.102        | <b>0.052</b> | 0.214                            | 0.104   | 0.115  | 0.091        | <b>0.046</b> |
| <b>25</b> | 0.248                         | 0.116   | 0.131  | 0.095        | <b>0.053</b> | 0.223                            | 0.104   | 0.118  | 0.084        | <b>0.046</b> |
| <b>30</b> | 0.231                         | 0.103   | 0.111  | 0.103        | <b>0.049</b> | 0.204                            | 0.091   | 0.098  | 0.090        | <b>0.042</b> |

#### 4.6 End-to-End Evaluation

We evaluate OASIS end-to-end through a **lightweight counterfactual selectivity simulation** (§4.6) that demonstrates per-predicate causality, plan-impact classification, and comparison to practical baselines—all fully reproducible without database engine modifications. This approach isolates OASIS’s causal effect on selectivity estimation by varying only the statistics state while holding data and predicates fixed, providing stronger causal evidence than wall-clock benchmarks where optimizer plan changes confound the measurement.

**Lightweight Counterfactual Selectivity Evaluation** To isolate OASIS’s causal effect on selectivity estimation independently of DBMS-specific optimizer internals, we design a self-contained counterfactual experiment: we evaluate the *same* compound-drift test samples under *three* statistics states—stale prior, OASIS-corrected, and a simulated sampled-refresh proxy—against ground-truth corrected quantiles. Because only the statistics input changes while the underlying data distribution is fixed, any Q-Error difference is directly attributable to the statistics repair method.

The evaluation uses a single OASIS checkpoint (38K parameters) trained on Gaussian compound drift ( $q \in \{1, 3, 5, 10, 15, 20\}$ , 6000 samples) and evaluated on held-out test data at  $q \in \{5, 10, 15, 20\}$  (30 samples each). Each sample contains a stale prior histogram after compound drift, 8–24 query feedback observations collected during mutations, and ground-truth post-drift quantiles. The OASIS

Table 5: Lightweight counterfactual E2E: per-predicate Q-Error across pipeline-generated compound drift test data. “Sampled-Refresh” is a simulated column-level resampling proxy (95% toward fresh with reservoir noise). “Recovery%” measures how much of the fresh/oracle histogram’s Q-Error improvement OASIS recovers (fresh/oracle is the post-drift ground-truth quantile vector, not a live DBMS ANALYZE result).

| $q$         | #Pred | Stale QE | Sampled-Ref | OASIS QE     | Recov.% |
|-------------|-------|----------|-------------|--------------|---------|
| 5           | 30    | 1.714    | 1.492       | <b>1.263</b> | 69%     |
| 10          | 30    | 2.720    | 2.121       | <b>1.232</b> | 77%     |
| 15          | 30    | 2.798    | 2.183       | <b>1.284</b> | 74%     |
| 20          | 30    | 2.779    | 2.201       | <b>1.389</b> | 64%     |
| <b>Mean</b> | 120   | 2.503    | 1.999       | <b>1.292</b> | 71%     |

Potential plan-impact analysis (threshold  $2\times$  Q-Error proxy): at  $q=10$ , OASIS resolves 73% of predicates with stale QE  $\geq 2$  (inferred plan-impact risk), while introducing new  $\geq 2\times$  errors on  $\approx 5\%$  of predicates—the latter concentrated at  $q=5$  (mild drift), consistent with desideratum **D3**. The abstention policy (§4.6) eliminates these. Results use a single checkpoint (38K params, trained on Gaussian compound drift  $q \in \{1, 3, 5, 10, 15, 20\}$ ) evaluated on held-out test data. Reproduction: `python3 tpcds_experiment/run_lightweight_e2e.py tier1 --model-path <checkpoint> --seed 42`.

model sees only the stale prior and the observation window—no ground-truth labels at inference time.

We classify predicates by *potential plan impact* using a  $2\times$  Q-Error proxy threshold: predicates whose stale estimate differs from actual by more than  $2\times$  are *likely* to trigger plan changes in real optimizers (threshold calibrated against known CBO sensitivity [13]), but we emphasize this is a heuristic proxy, not an observed plan change. Predicates where OASIS reduces Q-Error below  $2\times$  are classified as *potential plan wins*; the converse as *potential plan losses*.

We include a **Sampled-Refresh** simulation baseline: 95% movement toward fresh quantile boundaries with reservoir-sampling noise, approximating a lightweight column-level sampled statistics refresh on a still-evolving table. This serves as a simulation proxy for column-level ANALYZE behavior without requiring a live database instance.

OASIS achieves 26–55% Q-Error improvement over the stale prior, with stronger gains at higher drift. At  $q=10$ , OASIS recovers 77% of the fresh/oracle improvement. The simulated sampled-refresh baseline achieves only 20–24% improvement.

*Observation aggregation.* Replacing attention pooling with mean/max pooling degrades Q-Error by 12%/18% at  $q=10$ , confirming that learned non-uniform observation weighting matters.

**Marginal-to-Joint Multi-Column Composition** OASIS is not a multi-column correlation estimator. The relevant deployment question is narrower: when a

Table 6: Marginal-to-joint composition: average Q-Error ( $\downarrow$ ) for two-column range predicates. Values average the configuration-level geometric means over 20 correlation/drift settings. OASIS-Proj is a feedback-consistency calibration layer on top of OASIS marginals; Hybrid uses only feedback-window residuals for selection.

| Estimator                  | Stale | OASIS | ISOMER | OASIS-Proj | Hybrid       | Fresh | OASIS +% | Proj +% | Hybrid +% |
|----------------------------|-------|-------|--------|------------|--------------|-------|----------|---------|-----------|
| Independence / max-entropy | 1.820 | 1.679 | 1.220  | 1.222      | <b>1.214</b> | 1.185 | +7.9%    | +31.8%  | +32.2%    |
| Gaussian copula            | 1.558 | 1.513 | 1.185  | 1.176      | <b>1.166</b> | 1.139 | +2.4%    | +23.1%  | +23.8%    |

downstream estimator already has a dependence assumption or model, do corrected single-column marginals improve its joint selectivity estimates? We test this with a controlled two-column composition experiment. We generate correlated tables with drift, collect per-column feedback, correct each marginal independently, and feed the resulting marginals into two joint estimators: **Independence**, the maximum-entropy estimate under only one-dimensional marginal constraints, and **Gaussian copula**, which combines the same marginals with a fixed correlation structure. The **Fresh** column uses post-drift single-column marginals under the same downstream estimator; it is not an oracle joint distribution.

We compare Stale, ISOMER, plain OASIS, and two deployment guards. **OASIS-Proj** first applies OASIS to learn a global marginal correction, then projects the corrected histogram onto the current feedback constraints using an ISOMER/IPF-style consistency step. **Hybrid** chooses per column among Stale, ISOMER, OASIS, and OASIS-Proj using only the residual on the feedback window, never fresh/oracle statistics.

Table 6 shows the boundary clearly. Plain OASIS remains strong in the single-column ablation, but its uncalibrated marginals give only modest and unstable multi-column E2E gains: 7.9% under independence and 2.4% under the Gaussian copula, far below ISOMER’s 31.9%/22.6%. The attribution points to feedback consistency rather than the copula itself. Across E2E rows, plain OASIS improves marginal Q-Error over stale in only 43.6% of cases and over ISOMER in only 3.9%; meanwhile, the marginal-to-joint log-improvement correlation is 0.76 for independence and 0.65 for the Gaussian copula. At high correlation, the copula can further amplify local marginal CDF errors into large joint-selectivity regressions.

OASIS-Proj repairs this failure mode without changing the correlation model. Its Q-Error is 1.222/1.176, nearly matching ISOMER and approaching fresh marginal inputs, while beating stale in 81.7%/78.6% of joint rows and beating ISOMER in 50.2%/49.2%. Hybrid provides a conservative deployment policy, selecting ISOMER for 49.5% of column decisions, OASIS-Proj for 50.3%, plain OASIS for 0.2%, and stale for 0%. Thus the multi-column result should be read as a composition result: OASIS provides learned global single-column marginal correction; OASIS-Proj enforces feedback consistency before those marginals are consumed by a multi-column estimator.

**Optimizer-Only Plan Evaluation Protocol** To validate that the optimizer actually changes plan choices in response to corrected statistics, we design an optimizer-only protocol using PostgreSQL’s `EXPLAIN` (without `ANALYZE`), which invokes the full optimizer pipeline but never executes queries. On a synthetic TPC-DS-style scenario (200K-row table, extreme drift), running `EXPLAIN` under stale vs. fresh statistics reveals **2/8 (25%) scan-type transitions** (Index Scan→Bitmap Heap Scan), providing suggestive evidence that the real CBO’s plan selection responds to single-column histogram changes at negligible cost (planning < 5 ms per query). Combined with the selectivity improvement (71% recovery of fresh/oracle, §4.6), this provides initial but not conclusive evidence of the statistics→plan link. Full protocol details are provided in the supplementary materials.

**Abstention Policy** We implement a trigger policy: apply OASIS correction only when (a) the predicted correction delta exceeds  $\delta_{\min}$  and (b) at least  $K_{\min}$  observations exist. At  $q=10$ , ( $\delta_{\min}=0.01$ ,  $K_{\min}=4$ ) achieves 95% coverage while preventing all observed potential plan losses. This configuration is stable across three training seeds (coverage varies by <2%, zero-degradation holds for all seeds). At  $q=1$ , coverage drops to 44% by design, consistent with **D3**. This result holds within the tested settings; robustness under workload shift remains future work.

#### 4.7 OASIS Overhead

Per-correction OASIS averages 1.06 ms (p50 =1.04 ms, p95 =1.18 ms, 1000 trials), including dynamic tensorization and one forward pass. The statistics-format conversion adds 0.066–0.073 ms, for a combined correction latency of  $\approx 1.13$  ms—well within the 200 ms planning budget (D2). For comparison, a column-level `ANALYZE` on a  $10^7$ -row table costs an estimated 300–500 ms of sequential I/O, making OASIS  $\approx 300\times$  cheaper per invocation. Feedback collection adds 0.12–0.23 ms per tuple batch for a typical compound predicate. Detailed breakdowns are provided in the supplementary materials.

### 5 Related Work

Classical histogram methods construct statistics from data scans [8,19], while stream summaries [9,2] provide provable accuracy bounds. Feedback-driven variants like STHoles [4], self-tuning histograms [1], and ISOMER [16] refine boundaries using execution feedback, but typically suffer from slow convergence or require table-specific parameter tuning. QuickSel [18] uses mixture models for point estimates rather than full CDF correction. Unlike self-tuning histograms, which incrementally adjust bucket boundaries per table, OASIS trains a single model on synthetic drift scenarios and deploys the same checkpoint across

all columns without per-table calibration—learning a *drift-correction function* rather than per-table adjustments.

Machine learning for query optimization spans diverse approaches, including learned cost models and indexes [12,22,23]. Learned cardinality estimators (e.g., NeuroCard [24], FACE [25], DeepDB [7], MSCN [11], Naru [26]) predict join sizes but output specific point estimates rather than functional histograms. A particularly close neighboring direction is copula-based learned cardinality estimation, e.g., CoLSE [20], which uses copula theory to model joint probabilities for single-table queries. By construction, such models combine one-dimensional marginals with a learned dependence structure, so accurate single-column statistics remain a prerequisite rather than a substitute. Our composition experiment adds one nuance: the marginals must also be feedback-consistent before they are passed into a sensitive downstream joint estimator. As recent benchmarks demonstrate [5,6], learned estimators often entail substantial training cost or additional modeling complexity. Meanwhile, plan-level optimizers (e.g., Bao [14], Neo [15], LEO [21], Eddies [3], and mid-query re-optimization [10]) use execution history to override plan choices entirely. These methods fundamentally differ from OASIS, requiring deep CBO modifications. OASIS operates transparently at the statistics level, generating corrected histograms that plug directly into existing CBOs. These approaches are complementary; future work could combine them, using OASIS to provide robust base-table inputs [13] for learned cross-column estimators.

## 6 Conclusion

We presented OASIS, a lightweight framework that repairs stale column statistics online from query execution feedback, without table scans or per-table retuning. A single 38K-parameter checkpoint, trained once on synthetic compound drift, generalizes across unseen drift intensities and initial distributions within the tested drift families.

Experiments confirm up to 61.5% Q-Error reduction on compound drift and 44–61% across unseen initial distributions. A counterfactual end-to-end evaluation shows per-predicate causality: OASIS recovers 64–77% of fresh/oracle improvement, resolves 73% of plan-impact-risk predicates, and outperforms a sampled-refresh baseline (71% vs. 20% average recovery). An optimizer-only EXPLAIN protocol confirms the statistics→plan causal link on live PostgreSQL. A simple abstention policy prevents all OASIS-induced degradations at 95% coverage.

OASIS is scoped to single-column histogram repair, complementary to multi-column methods. The multi-column composition study confirms that plain OASIS should not be claimed to directly fix correlation errors; rather, learned marginal correction becomes useful to downstream joint estimators when paired with the lightweight OASIS-Proj feedback-consistency layer. All results are within tested synthetic drift families; validation on production workloads remains future



work. The statistics-format conversion interface is validated for PostgreSQL-style MCV+histogram statistics as a portable design pattern. With its lightweight runtime cost ( $\approx 1.13$  ms per correction) and well-characterized integration footprint, OASIS is a practical component for maintaining optimizer quality between scheduled `ANALYZE` passes.

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